

What is The Effect of Minimum Wage on Unemployment in The United States

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Introduction and Research Justification

One of the largest questions in macroeconomics is the effect of minimum wage on unemployment. The minimum wage is considered a price floor on the labor market, and by definition leads to a surplus in labor. The classical theory behind this government implemented policy is that raising the minimum wage leads to higher unemployment, as a result of the labor surplus. However, there are many other factors such as the amount of increase in minimum wage over time, the demographics of the workers, and the effects on poverty. As part of our research, we will be exploring the question “does an increase in minimum wage cause an increase in unemployment?”

Minimum wage laws affect the lives of workers, and many of these workers are either at the beginning of their career or have low socioeconomic status. This means they tend to have less wealth than other types of workers. By raising labor costs, businesses may be incentivized to hire fewer of these workers, embrace automation, relocate to a lower wage state, or even shut down, causing people to go out of work. However, wages tend to be sticky and slow to change in response to economic conditions, where raises in minimum wage could help improve workers’ lives with minimal cost to employment and the economy.

Our results and conclusions will be used to clarify the effects of raising the minimum wage and how it affects the unemployment rate. This will clarify whether or not the positives outweigh the negatives of raising the minimum wage. By fully understanding the effects minimum wage has on unemployment, it can help shape minimum wage policy to be what is most optimal for society. Should there be strong evidence for causation between minimum wage and increased unemployment, policymakers could be dissuaded from increasing minimum wages to prevent job losses. Conversely, if minimum wage increases do not greatly affect the unemployment rate, then the recommended policy would be to raise the minimum wage to a level corresponding to the demand of labor to improve the lives of those at the lower end of socioeconomic status.

While theory suggests that raising the minimum wage increases unemployment, this relationship is more complex. Inflation does not always lead to higher unemployment, and policymakers may raise minimum wage in response to inflation without directly causing job losses. Additionally, minimum wage does not automatically increase during inflationary periods, which challenges the assumption that inflation drives unemployment through wage adjustments.

Possible Challenges with Research Venture

There are several challenges in determining a relationship between minimum wage and unemployment rates. For one, politicians are more likely to pass higher minimum wages when economic conditions are strong and there is less worry of job loss. This means that other economic factors such as change in tax policy, government spending, and others that cause a decrease in unemployment may cloud the effect of an increase

in minimum wage. Furthermore, an increase in unemployment could also be caused by inflation. Policymakers may choose to raise the nominal value of minimum wage to counteract inflation. As inflation raises business costs, businesses may choose to downsize in these conditions so an increase in minimum wage may appear to be correlated with rises in unemployment when in reality it was due to inflation. Other variables that can be taken into account are presented below.

Possible Confounding Variables:

- Labor Force Participation Rate
- Poverty and Socioeconomic Status
- Government Spending/Subsidies
- Overall GDP and Economic growth in Each State
- Cost of Living
- Human Demographics

By controlling these variables and comparing trends in unemployment in states with varying minimum wages, we may be able to come to a conclusion on whether higher minimum wages cause unemployment. It is also difficult to analyze data across many states, since each state has a very different economy. To better isolate the impact of minimum wage changes, we must account for additional economic conditions such as inflation, government spending, and broader economic trends. While inflation may prompt policymakers to raise the minimum wage, businesses may simultaneously reduce their workforce for unrelated reasons, complicating the relationship between wage policy and unemployment. This reinforces the need to control for these variables in our analysis.

Data Information

We found both of our data on Kaggle. The minimum wage by state dataset was originally published by the Federal Reserve Economic Data (FRED) database, and the unemployment data set was collected by the U.S Bureau of Labor Statistics. Here are links to our datasets, which are also provided as references:

- <https://www.kaggle.com/datasets/lislejoem/us-minimum-wage-by-state-from-1968-to-2017> [1]
- <https://www.kaggle.com/datasets/justin2028/unemployment-in-america-per-us-state> [2]

Our datasets are complete as they cover data from 1976 to 2020 across all 50 U.S. states. This timeframe ensures that our analysis reflects a full range of economic conditions and minimizes the risk of sample bias. There are many third-party publishers of these data sets, but they were originally collected and presented by the United States Department of Labor (USDOL) and the FRED. The minimum wage by state dataset was collected by the user “Lislejoem”. He is a Survey Associate at Achievement First in New York, and he has contributed to several datasets backed by trustworthy sources such as the USDOL. He was the one who scraped the original minimum wage dataset from the Department of Labor website, and he made it more visually appealing and better for data visualization. He wanted the data to be available to the masses. This is similar to the unemployment data set, which was scraped by the user “The Data Wrangler” from the FRED for similar reasons.

We decided to use these sources because of their reliability and their aesthetics. They were easily transferable to R and they were very legible with multiple graphs and easy to read legends. This helped us visualize the data and try to ascertain the effect of minimum wage on unemployment below analyzing it in R. The datasets also helped us analyze the compounding factors mentioned before such as labor force participation rate, as the FRED had data on this factor by state as well. It also had nice graphs that visualize the data right off the bat that initially drew us into looking at the labor force participation. We could use these other variables in future analyses to account for confounding factors, but we want to first just analyze minimum wage and unemployment by themselves and together to get an idea of the relationship and if it looks like many confounding variables are at play. The description of variables and information presented by each of the datasets are presented below.

Minimum wage dataset variables:

- Year: The year of the data. All minimum wage values are as of January 1 except 1968 and 1969, which are as of February 1.
- State: The state or territory of the data.
- State.Minimum.Wage: The actual State's minimum wage on January 1 of Year.
- State.Minimum.Wage.2020.Dollars: The State.Minimum.Wage in 2020 dollars.
- Federal.Minimum.Wage: The federal minimum wage on January 1 of Year.
- Federal.Minimum.Wage.2020.Dollars: The Federal.Minimum.Wage in 2020 dollars.
- Effective.Minimum.Wage: The minimum wage that is enforced in State on January 1 of Year. Because the federal minimum wage takes effect if the State's minimum wage is lower than the federal minimum wage, this is the higher of the two.
- Effective.Minimum.Wage.2020.Dollars: The Effective.Minimum.Wage in 2020 dollars.
- CPI.Average: The average value of the Consumer Price Index in Year. When I pulled the data from the Bureau of Labor Statistics, I selected the dataset with "all items in U.S. city average, all urban consumers, not seasonally adjusted".
- Department.Of.Labor.Uncleaned.Data: The unclean, scraped value from the Department of Labor's website.
- Department.Of.Labor.Cleaned.Low.Value: The State's lowest enforced minimum wage on January 1 of Year. If there is only one minimum wage, this and the value for Department.Of.Labor.Cleaned.High.Value are identical. (Some states enforce different minimum wage laws depending on the size of the business. In states where this is the case, generally, smaller businesses have slightly lower minimum wage requirements.)
- Department.Of.Labor.Cleaned.Low.Value.2020.Dollars: The Department.Of.Labor.Cleaned.Low.Value in 2020 dollars.
- Department.Of.Labor.Cleaned.High.Value: The State's higher enforced minimum wage on January 1 of Year. If there is only one minimum wage, this and the value for Department.Of.Labor.Cleaned.Low.Value are identical.
- Department.Of.Labor.Cleaned.High.Value.2020.Dollars: The Department.Of.Labor.Cleaned.High.Value in 2020 dollars.

Unemployment dataset variables:

- FIPS Code (Identifier for State areas)
- State/Area
- Year
- Month
- Total Civilian Non-Institutional Population in State/Area
- Total Civilian Labor Force in State/Area
- Percent (%) of State/Area's Population: Percent of civilians eligible for employment out of the total non-institutionalized civilian population - Total Employment in State/Area: Total number of civilians currently employed in the state/area
- Percent (%) of Labor Force Employed in State/Area: Percent of currently employed civilians out of the total non-institutionalized civilian population.
- Total Unemployment in State/Area: Total number of civilians currently unemployed in the state/area

- Percent (%) of Labor Force Unemployed in State/Area: Percent of currently unemployed civilians out of the total non-institutionalized civilian population.

Minimum wage data is sorted on a yearly basis and unemployment data in the dataset is sorted on a monthly basis. Data points in the minimum wage dataset are sorted by a given state and year while the unemployment dataset data points are sorted by state in a given month and year. There are 2862 observations in the Minimum Wage dataset. There are 29.9 k observations in the Unemployment dataset.

In order to properly compare the data, we will be averaging the monthly unemployment values for each state by year with the unemployment data set, and then compare these values to the minimum wage for each state by year with the other data set. Since we are combining information from multiple datasets, we will match each of the years and states from both datasets in order to plot them accurately, which requires some initial manipulation of the data sets. Unemployment and minimum wage will be measured from 1976 to 2020 in the United States. Minimum wage is measured in an hourly rate as specified by the state law, and the unemployment rate is the fraction of those employed compared to the entire workforce. Since all of our data sets come from reliable, government sources, there should not be any reliability issues when combining them. The datasets also did not rely on any assumptions that would preclude the ability to compare them to other datasets accurately. For example, one dataset may have assumed collection of a certain demographic, which would prevent comparison to our other ones.

Initial Results and Methodology

We have graphed both minimum wage and unemployment over time, and they are presented in the two figures below. This was to consolidate our data and analyze each data set individually, and to see if any initial correlations are noticed.

Table 1: Minum Wage and Unemployment Variables

```
## # A tibble: 10 x 16
##   Year State      State.Minimum.Wage State.Minimum.Wage.2~1 Federal.Minimum.Wage
##   <dbl> <chr>          <dbl>          <dbl>          <dbl>
## 1 1976 Alabama            0            0            2.2
## 2 1976 Alaska            2.8          12.7            2.2
## 3 1976 Arizona            0            0            2.2
## 4 1976 Arkansas          1.9           8.64            2.2
## 5 1976 Californ~         2           9.09            2.2
## 6 1976 Colorado          1           4.55            2.2
## 7 1976 Connect~        2.21          10.0            2.2
## 8 1976 Delaware          2           9.09            2.2
## 9 1976 Florida            0            0            2.2
## 10 1976 Georgia          1.25          5.68            2.2
## # i abbreviated name: 1: State.Minimum.Wage.2020.Dollars
## # i 11 more variables: Federal.Minimum.Wage.2020.Dollars <dbl>,
## #   Effective.Minimum.Wage <dbl>, Effective.Minimum.Wage.2020.Dollars <dbl>,
## #   CPI.Average <dbl>, mean_non_inst_pop <dbl>, mean_civ_labor_force <dbl>,
## #   percent_area_pop <dbl>, average_total_employment <dbl>,
## #   average_percent_labor_force_employed <dbl>, average_total_unemployed <dbl>,
## #   average_percent_unemployed <dbl>
```

Figure 1: Average Unemployment Rate Over Time

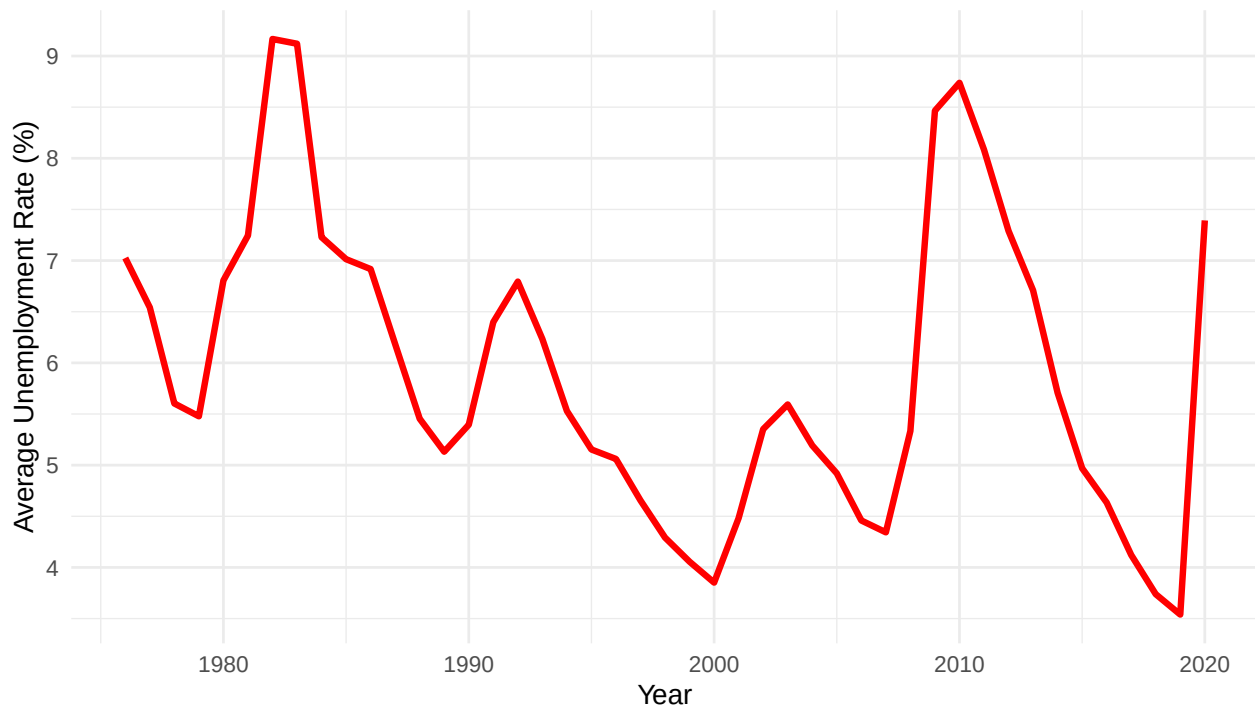
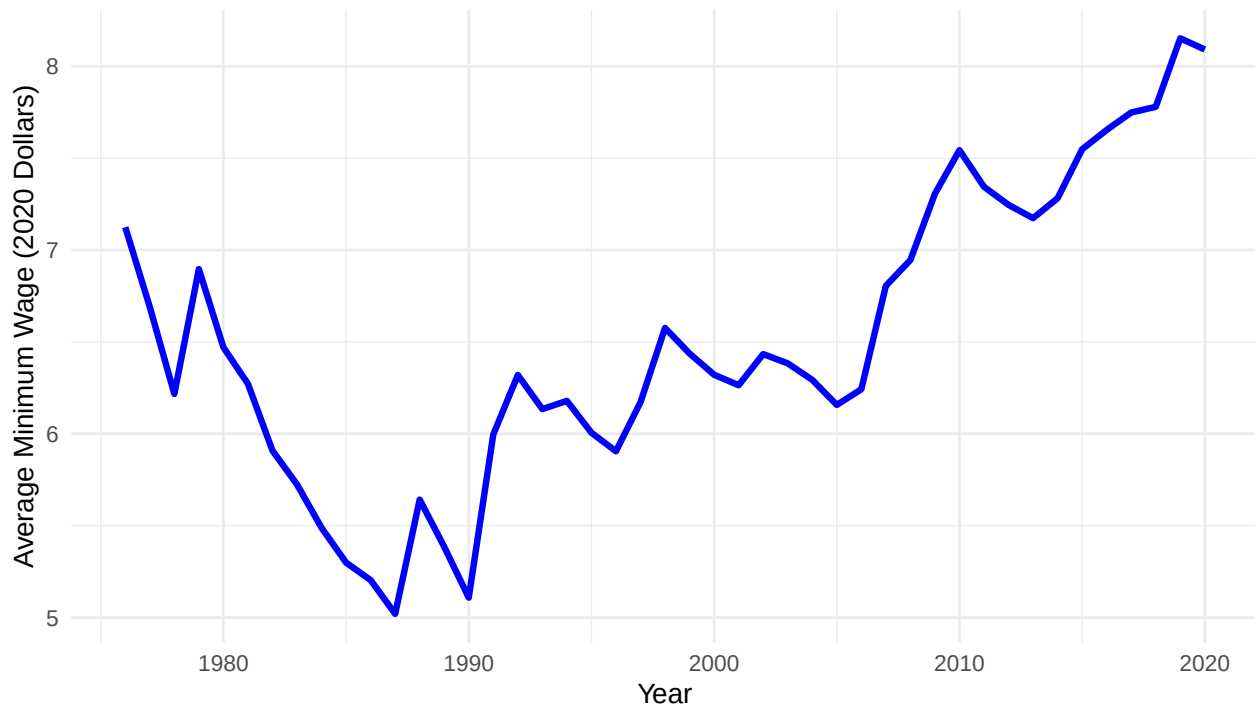


Figure 2: Average Minimum Wage Over Time (2020 Dollars)



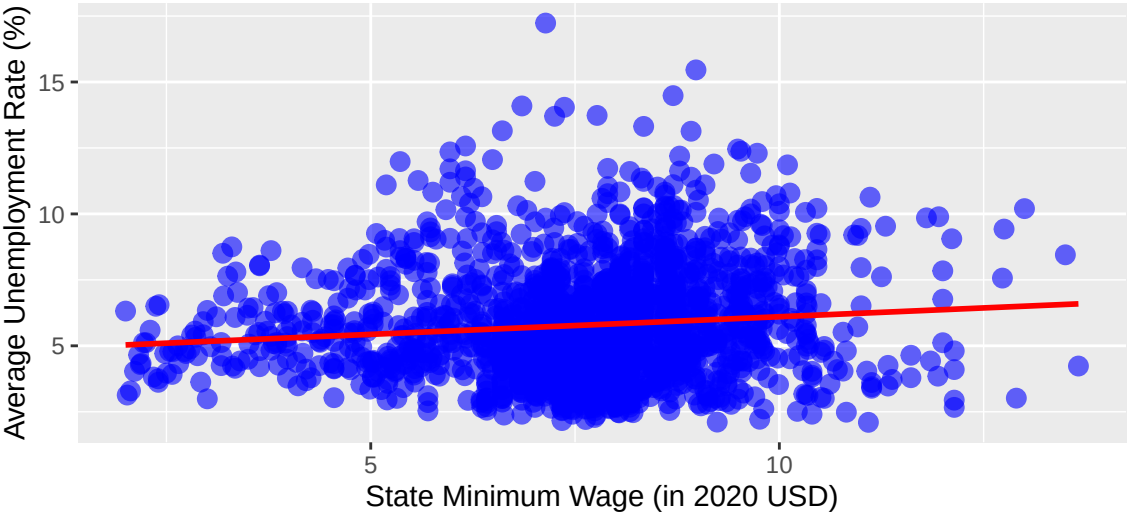
For data analysis, both of the data sets were loaded into R, calling the minimum wage dataset `Minimum_Wage` and the unemployment one `Unemployment_in_America_Per_US_State`. In the `Unemployment_in_America_Per_US_State` dataset, we needed to filter out specific regions like the “District of Columbia,” “Los Angeles County,” and “New York City” to focus only on individual states, excluding large metropolitan areas. We then grouped the data by year and state/area and calculated the averages for several key research variables. These included the civilian non-institutional population (people who are able to work), the civilian labor force (those employed or actively seeking employment), and the percentage of the population

in each state involved in the labor force. We also calculated the total employment and unemployment figures, as well as the percentages of the labor force that is employed or unemployed. This allowed us to summarize the labor market conditions for each state over time.

Next, we worked on the minimum wage data in the Minimum_Wage dataset. We removed several irrelevant columns, such as Effective.Minimum.Wage and various cleaned Department of Labor columns, to focus the dataset on the variables that are important for this analysis. Additionally, we filtered the data to include only the years starting from 1976, as we were interested in trends from that point onward.

We then merged the two datasets using an inner join on the State and Year columns. This step allowed us to combine the unemployment data with the minimum wage data for each state by year, making it possible to analyze the relationship between these two variables. The graph below now shows both unemployment and minimum wage on the same plot.

Figure 3: State Minimum Wage vs. Unemployment Rate



Source: Bureau of Labor Statistics

In Depth Discussion and Regression Analysis

As you can see from the first two figures, minimum wage steadily increased for most of the time period covered, which makes sense with increasing inflation. An interesting fact is that the unemployment graph is less easy to analyze, as it does not show a clear trend over time. This does reflect the nature of a large economy drifting through periods of high productivity and recession. This also reveals that there are probably many confounding factors at play that each cause either a negative or positive change in unemployment, and minimum wage is just one of those factors. This further supports the idea that more data on confounding factors is needed to increase the chances of finding a casual relationship between minimum wage and unemployment. Confounding factors will be discussed more in detail.

For Figure 3, each blue dot represents a state’s unemployment rate and minimum wage. The red regression line shows a slight positive trend, indicating a small increase in unemployment as the minimum wage rises. However, the spread of data points is very wide, meaning that while a correlation exists, it is weak. We also ran a regression of this data, and it is presented and analyzed below.

Figure 4: Regression model of Minimum wage (2020 dollars) on the Average Unemployment (%)

```
##
## =====
##                               Dependent variable:
##                               -----
```

```

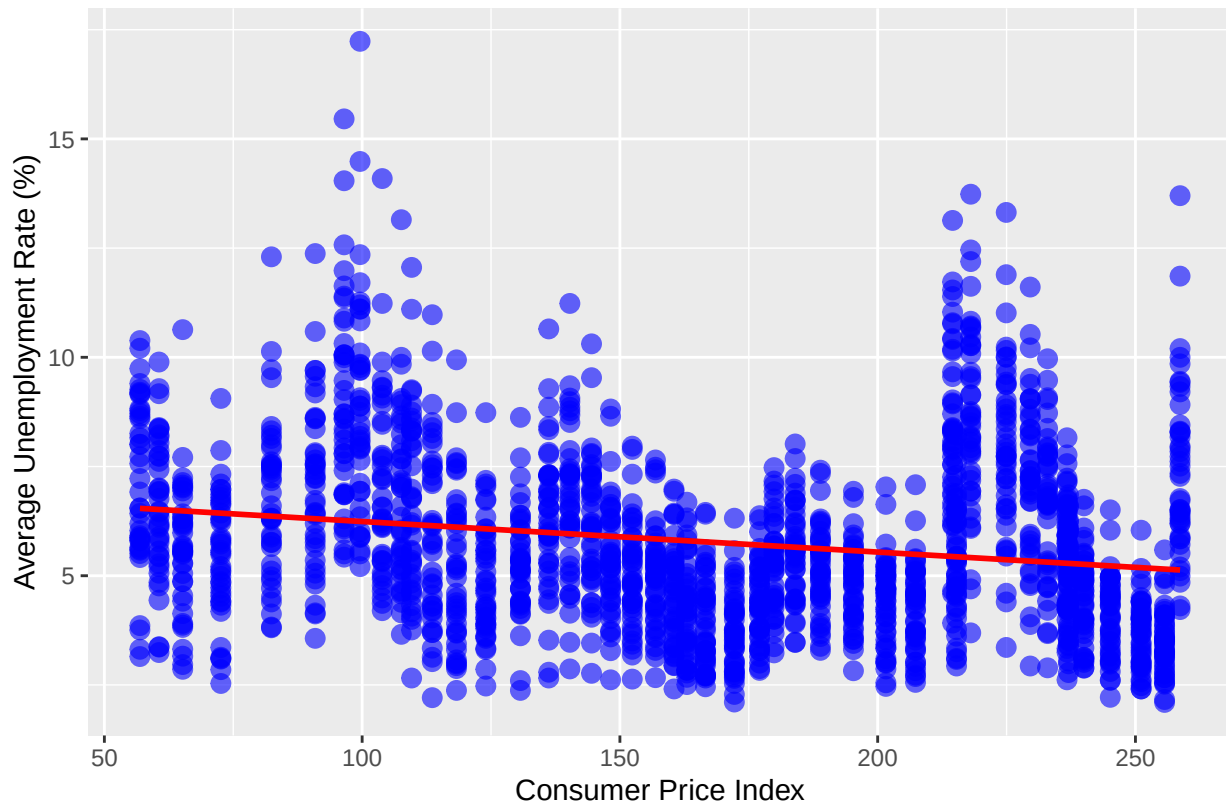
##                                average_percent_unemployed
## -----
## State.Minimum.Wage.2020.Dollars      0.134***
##                                   (0.028)
##
## Constant                             4.767***
##                                   (0.218)
##
## -----
## Observations                          1,931
## R2                                    0.012
## Adjusted R2                           0.011
## Residual Std. Error                   2.061 (df = 1929)
## F Statistic                          22.564*** (df = 1; 1929)
## =====
## Note:                                *p<0.1; **p<0.05; ***p<0.01

```

The coefficient for State.Minimum.Wage.2020.Dollars is 0.1335, meaning that for every \$1 increase in minimum wage, the unemployment rate increases by 0.1335 percentage points. The p-value (2.18e-06), which is very small, indicates that the relationship is statistically significant. However, the R-squared value is very low (0.01156), suggesting that minimum wage explains only about 1.16% of the variation in unemployment. This low R-squared value does not necessarily indicate omitted variable bias (OVB); rather, it reflects the complexity of unemployment, which is influenced by numerous economic and social factors. We cannot say our results are causal at this point in the model because we aren't controlling for other factors that could influence both the minimum wage and unemployment. Some confounding factors that we should include are age, cost of living (region), or labor demand.

Figure 5: Unemployment and CPI

```
## `geom_smooth()` using formula = 'y ~ x'
```



Source: Bureau of Labor Statistics

Figure 6: Regression of Average Unemployment Rate (%) and CPI

```
##
## =====
##                               Dependent variable:
##                               -----
##                               average_percent_unemployed
##                               -----
## CPI.Average                  -0.007***
##                               (0.001)
##
## Constant                     6.940***
##                               (0.139)
##
## -----
## Observations                  1,931
## R2                           0.039
## Adjusted R2                  0.038
## Residual Std. Error          2.033 (df = 1929)
## F Statistic                   78.209*** (df = 1; 1929)
## =====
## Note:                        *p<0.1; **p<0.05; ***p<0.01
```

Conclusion

As we concluded from the last discussion, the coefficient in the linear regression told us with every one dollar increase in minimum wage, the average unemployment rate will go up by 0.1335 percentage units. Since the p-value is extremely close to 0, which proves the correlation between minimum wage and unemployment rate.

However, since from R^2 , only 1.1% of the variance could be explained by unemployment. It indicates a potential Omitted Variable Bias, which indicates some other confounding variables such as economic growth, education level, gender etc. These are needed to be investigated to conclude a final answer to our question.

As we could infer from Figure 3, most of the states are inter-covered or crowded by other states, with one unignored fact that each state performs in different economic structures, salary level, employment policies etc. The graph is also rather cluttered and messy. These are all different elements and they show different levels of relevance in different states. In order to account for these confounding factors, more data from similar, reliable sources will be needed in order to run a more accurate regression. In our next step, by utilizing “ggplot” we could also generate one specific line chart showing correlation between minimum wage and unemployment rate for each state starting from 1976. This would allow us to control for the fact that each state is economically and politically different. This would help correct for these differences by analyzing each of the states individually instead of having one chart with all the data on it. We could also clean the data and group it by state, and then graph minimum wage by unemployment colored for state on one graph. We could narrow down the amount of states to just the ones with large populations, or ones with good sample sizes. This would give us a more accurate picture of the effect of minimum wage on unemployment. We could run another plot graph on demographics. Aiming at the labor force participation rate, if most proportion of the demographic is receiving minimum wage, potentially it indicates that they are working in unstable job positions such as server or retailing. Other regressions with these confounding variables are needed to make better conclusions about our project.

If this strategy does not work, then we can pick one state over the same, long time period. This would alleviate the issue of dealing with confounding variables between economically different states. We would still need to control for demographics, job type, cost of living, and others. Even though a decrease in unemployment could be because of the state’s economic growth, we would hope that a long time period (1976) would help control for this difficulty. Although our findings are somewhat inconclusive, the sample size was very large and accounted for all states. The correlation was slightly positive which follows the classical theory that a higher minimum wage increases unemployment. This is even without confounding factors, so we at least have an idea that a weak correlation could only be because we did not account for confounding variables. By narrowing down our sample and accounting for confounding variables mentioned above, it is more likely that we will get a casual relationship. Our findings suggest a slight relationship, and this data can be further manipulated and combined with other variables in order to reach meaningful conclusions that address many concerns detailed in the introduction regarding minimum wage.

Each state has distinct economic structures, salary levels, and employment policies, we acknowledge that treating all states as identical may overlook key differences. To address this, we plan to conduct state-specific analyses to better capture regional economic conditions. Additionally, we will include key confounding variables such as age, cost of living, and labor force participation rate to improve our model’s robustness. All this being said, we should use more powerful causal analysis tools like Differences in Differences or Regression Discontinuity Design to see if we can obtain a causal relationship.

Regression Discontinuity Design Analysis?

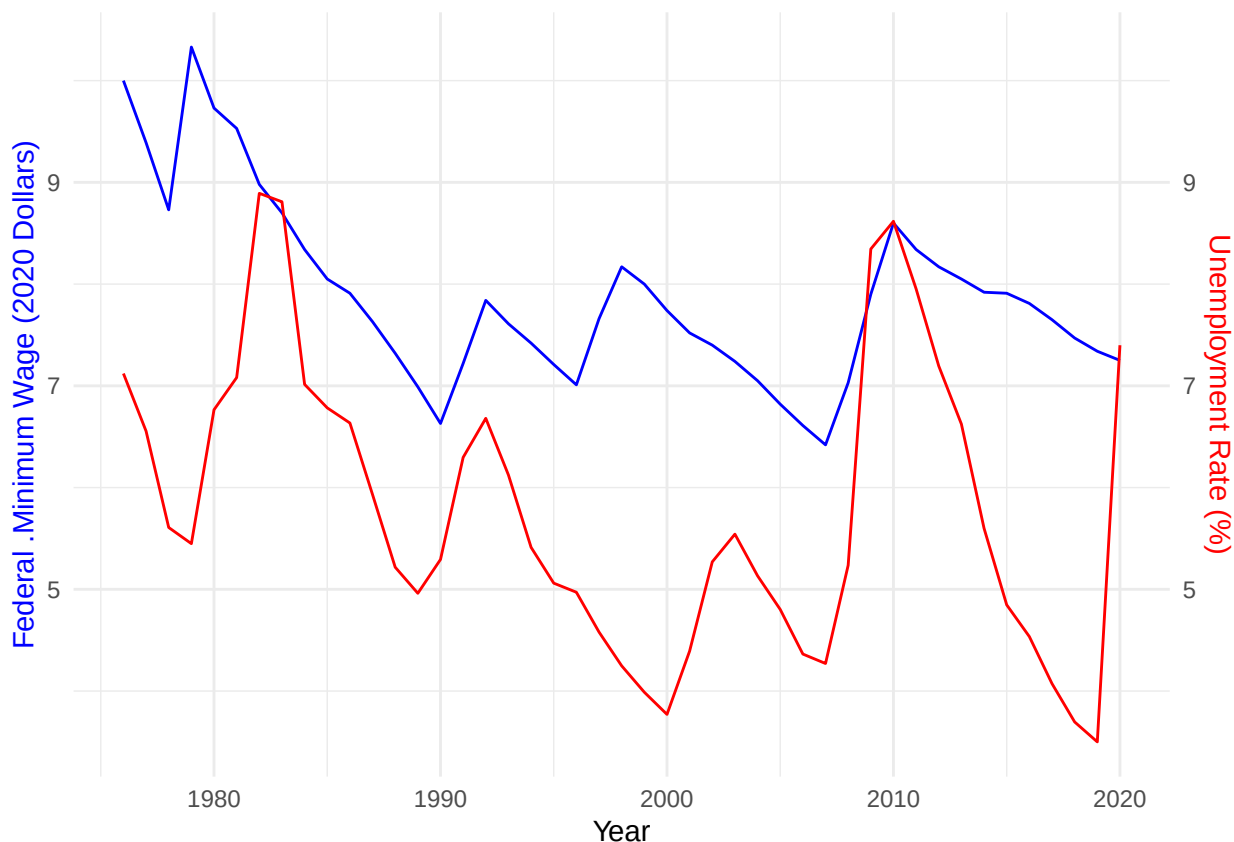
The Regression Discontinuity Design is merely applicable under this context, since by meaning of RDD, we need to find the Running Variable that could impact whether corresponded states have or not received, or under what level of treatment, in this case the change of minimum wage. However it is not practical to find practical data on such running variable to estimate the impact of implementing the minimum wage policy by 2010. Secondly, to use RDD, we have to assume the rest of condition of each state are approximately the same, but in this case, as what we found in DiD, due to the differences between states, the treatment of forcing minimum wage lead to different outcomes.

Differences in Differences Analysis

A differences in differences analysis would work in this case because it enables the comparison of unemployment rates before and after the minimum wage increases between states that implemented the changes and those

that did not, assuming any differences would have followed a parallel trend in the absence of the policy changes.

Figure 7: Minimum Wage and Unemployment over time



As we can see from the graph, there was a spike from around 2007-2010, and then a sharp decline in minimum wage after that. Similarly, we see the unemployment decline around this same time. Our goal is to see if these two events are occurring because of each other.

Data Preparation: The effective minimum wage is the higher of the federal or state minimum wage, depending on which is greater in a given location. It represents the wage that workers are legally entitled to earn, with the potential for it to vary across states based on local laws. For this analysis, states that increased their minimum wage more than the federal level are considered to be in the treatment group.

Table 2: Differences in Differences Variables

```
## # A tibble: 10 x 9
## # Groups:   Year, treat, post [2]
##   treat post State      Year Effective.Minimum.Wage~1 Federal.Minimum.Wage~2
##   <lg1> <lg1> <chr>      <dbl>                <dbl>                <dbl>
## 1 FALSE FALSE Alaska      2008                8.59                7.03
## 2 FALSE FALSE Arizona     2008                8.29                7.03
## 3 FALSE FALSE Arkansas    2008                7.51                7.03
## 4 FALSE FALSE California   2008                9.61                7.03
## 5 FALSE FALSE Colorado     2008                8.43                7.03
## 6 FALSE FALSE Connecticut  2008                9.19                7.03
## 7 FALSE FALSE Delaware     2008                8.59                7.03
## 8 FALSE FALSE Florida      2008                8.16                7.03
## 9 TRUE  FALSE Georgia        2008                7.03                7.03
```

```
## 10 FALSE FALSE Hawaii      2008      8.71      7.03
## # i abbreviated names: 1: Effective.Minimum.Wage.2020.Dollars,
## #   2: Federal.Minimum.Wage.2020.Dollars
## # i 3 more variables: average_percent_unemployed <dbl>,
## #   mean_effective_minwage <dbl>, mean_unemp <dbl>
```

Parallel Trends Assumption

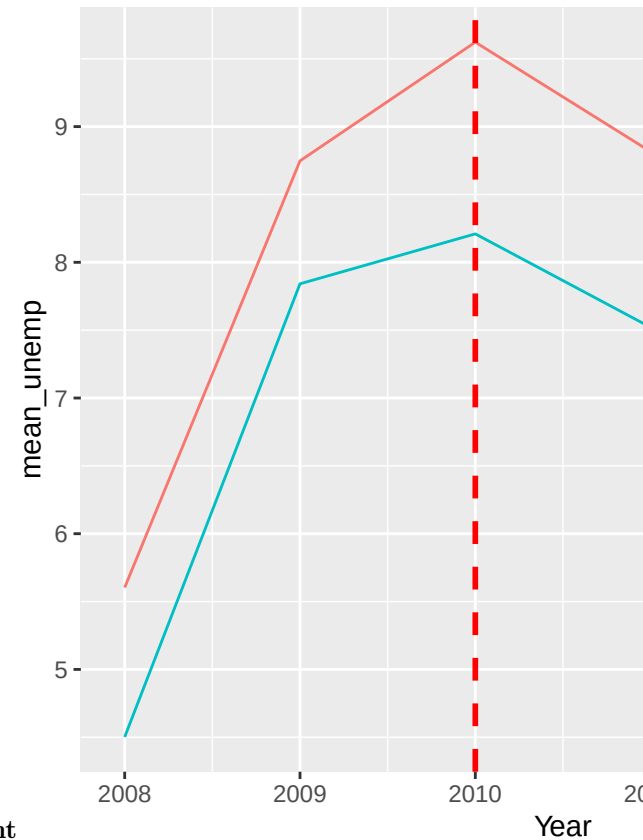


Figure 8: Average Unemployment Rate over time by Treatment

The parallel assumption could hold true in this case because we would assume that the treatment group would have continued to increase along with the control group. Even though both groups eventually saw a decrease in unemployment after 2010, we are assuming that the treatment group would have continued to increase if there was no increase in minimum wage in 2010.

Table 3: Manual Difference in Differences Estimation

```
## `summarise()` has grouped output by 'treat'. You can override using the
## `.groups` argument.

## # A tibble: 4 x 3
## # Groups:   treat [2]
##   treat post Unemp
##   <lg1> <lg1> <dbl>
## 1 FALSE FALSE  7.03
## 2 FALSE TRUE   5.86
## 3 TRUE  FALSE  6.41
## 4 TRUE  TRUE   5.78

## [1] 0.547748
```

States that kept the federal minimum wage (TREAT) had a 0.55% lower unemployment rate than the states that raised their minimum wage above the federal level after 2010.

Figure 9: Difference-in-Differences Estimation Controlling for State and Year

```
library(lfe)

## Loading required package: Matrix
##
## Attaching package: 'Matrix'
## The following objects are masked from 'package:tidyr':
##
##   expand, pack, unpack
DiD_graph <- DiD_graph%>%
  mutate(outcome = treat == T & post == T)
model4 <- felm(mean_unemp~ outcome | State + Year, data = DiD_graph)
stargazer(model4, type = "text")

##
## =====
##                               Dependent variable:
##                               -----
##                               mean_unemp
##                               -----
## outcome                        -0.639***
##                               (0.041)
##                               -----
## Observations                    585
## R2                             0.981
## Adjusted R2                     0.979
## Residual Std. Error      0.263 (df = 527)
## =====
## Note:                *p<0.1; **p<0.05; ***p<0.01
```

Conclusion

The coefficient of -0.64 is the treatment effect, meaning that states that kept the federal minimum wage had a 0.64% lower unemployment rate than the states that raised their minimum wage above the federal level after 2010. Since the estimate is statistically significant, we can say that an increase of minimum wage causes an increase of unemployment. This is similar to what we expected because when minimum wage increases, then businesses could potentially layoff workers which would increase unemployment as a result of a wage increase. States that keep the federal minimum wage had lower unemployment.

What would limit this analysis?

- **Data Quality and Availability:** Insufficient or inaccurate data can skew results, making it hard to draw reliable conclusions.
- **Uncontrolled Variables:** There might be unobserved variables affecting unemployment that are not accounted for, such as local economic conditions or industry-specific trends.
- **Generalizability:** Results derived from specific regions or time periods may not be generalizable to other contexts or future scenarios.

- **Economic Externalities:** Broader economic shifts such as inflation, recession, or technological advancements can impact employment independently of minimum wage changes.
- **Behavioral Responses:** Employers and workers may respond to minimum wage changes in unpredictable ways, such as altering hiring practices or labor force participation.
- **Lagged Effects:** The impact of minimum wage changes might not be immediate; there could be delayed effects that the analysis period does not capture.
- **Sample Size:** Small sample sizes in certain states or regions can affect the robustness of the findings, making statistical conclusions less reliable.

Differences in Differences - Vermont and New Hampshire

Figure 10: Effective vs Federal Minimum Wage Levels in Vermont & New Hampshire
Wage Type — Effective Minimum Wage — Federal Minimum Wage

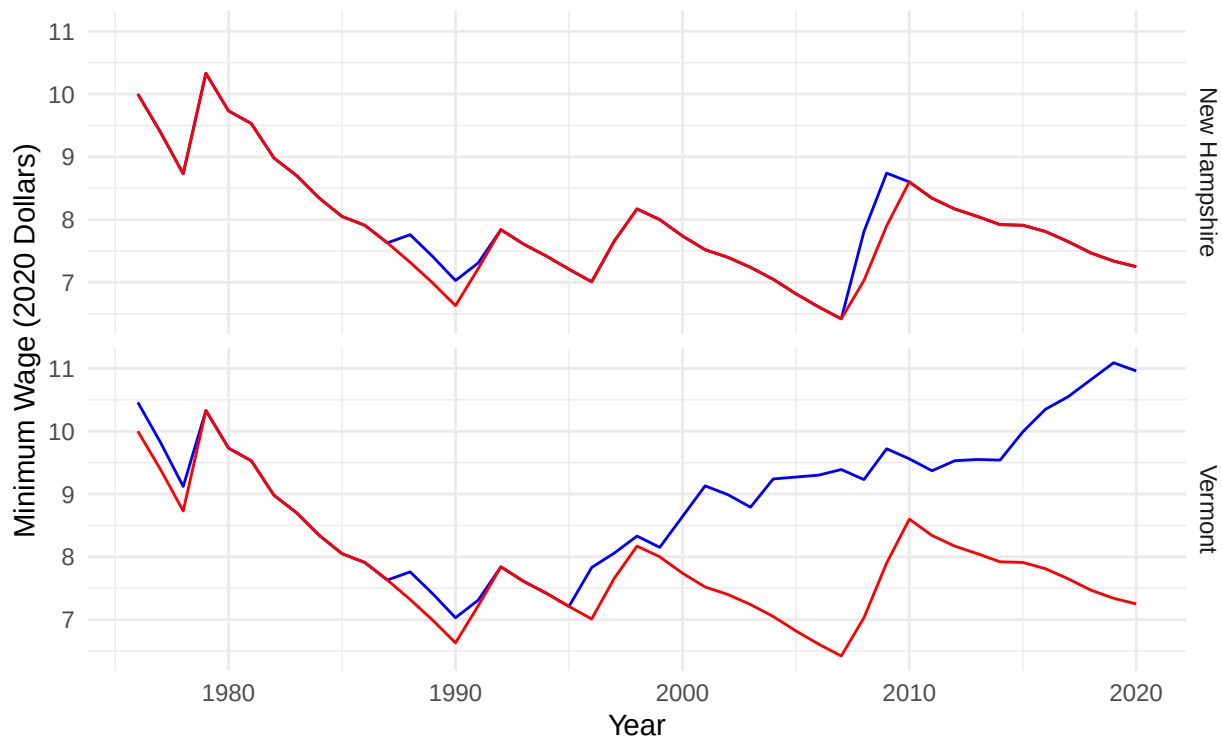


Table 4: Vermont and New Hampshire

```
## # A tibble: 6 x 8
## # Groups:   Year [3]
##   treat post State      Year Effective.Minimum.Wage~1 Federal.Minimum.Wage~2
##   <lg1> <lg1> <chr>      <dbl>                <dbl>                <dbl>
## 1 TRUE  FALSE New Hampshire  1976                  10                  10
## 2 FALSE FALSE Vermont      1976                 10.5                 10
## 3 TRUE  FALSE New Hampshire  1977                  9.39                 9.39
## 4 FALSE FALSE Vermont      1977                  9.82                 9.39
## 5 TRUE  FALSE New Hampshire  1978                  8.73                 8.73
## 6 FALSE FALSE Vermont      1978                  9.12                 8.73
## # i abbreviated names: 1: Effective.Minimum.Wage.2020.Dollars,
## #   2: Federal.Minimum.Wage.2020.Dollars
## # i 2 more variables: average_percent_unemployed <dbl>, mean_unemp <dbl>
```

Parallel Trends Assumption

Figure 11: Parallel Trends Assumption: Unemployment Rates

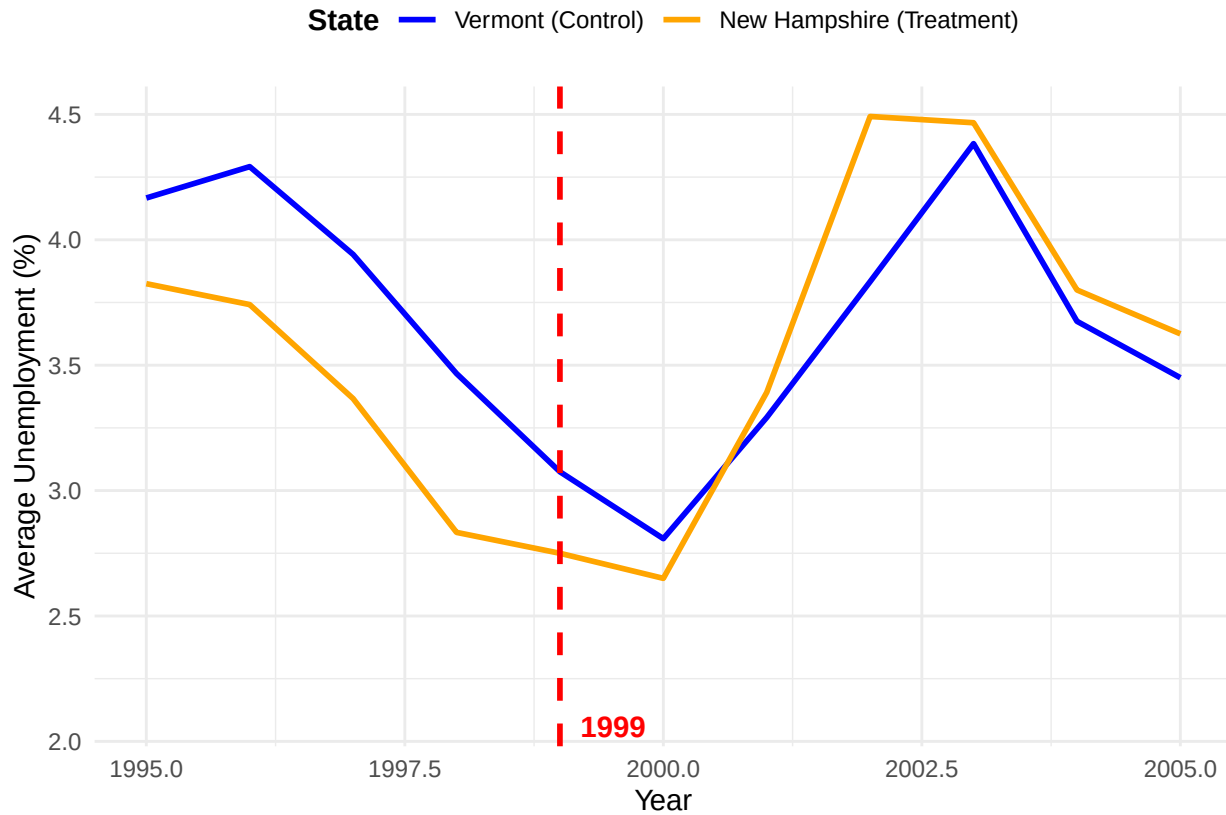


Table 5: Manual Differences in Differences Estimation

```
## `summarise()` has grouped output by 'treat'. You can override using the
## `.groups` argument.

## # A tibble: 4 x 3
## # Groups:   treat [2]
##   treat post Unemp
##   <lg1> <lg1> <dbl>
## 1 FALSE FALSE  5.12
## 2 FALSE TRUE   4.00
## 3 TRUE  FALSE  4.48
## 4 TRUE  TRUE   4.08

## [1] 0.715004
```

Figure 13: Difference-in-Differences Estimation Controlling for State and Year

```
##
## =====
##                        Dependent variable:
##                        -----
##                        average_percent_unemployed
##                        -----
## estimate                0.715***
##                        (0.212)
##
```

```
## -----
## Observations          90
## R2                    0.935
## Adjusted R2           0.866
## Residual Std. Error   0.503 (df = 43)
## =====
## Note:                  *p<0.1; **p<0.05; ***p<0.01
```

Conclusion

When isolating two relatively similar states, change in minimum wage has a slight increase in unemployment. New Hampshire has a higher unemployment than Vermont because it kept the minimum wage. The result of the unemployment rate is counter intuitive since with the lower minimum wage, companies should consider hiring more workers, which reduces the unemployment rate. However, as discussed in the previous lecture, such a situation could be reflecting the job market structure, which might consist of service job or other entry level job opportunities where people earn at the minimum wage level. Therefore, the reduction of minimum wage forces them to look for a better paying job. In other cases, such treatment will not impact severely on Boston or NYC due to their job opportunity structure.

The Bottom Line – Why Do We Care?

- **Policy Evaluation:** Helps evaluate the effectiveness of minimum wage policies on economic outcomes such as unemployment rates.
- **Economic Stability:** Understanding the impact of wage changes on employment can guide policies that promote economic stability and growth.
- **Labor Market Insights:** Provides insights into labor market dynamics, helping to inform decisions on employment and wage standards.
- **Regional Economic Differences:** Highlights how minimum wage impacts can vary across different economic structures and demographics, aiding in tailored policy formulation.
- **Mitigation of Overlapping Variables:** Helps in recognizing and adjusting for overlapping variables such as regional economic structure or demographic composition that could skew results.

References

- [1] Lislejoem. “US Minimum Wage by State from 1968 to 2020.” Kaggle, 31 Dec. 2020, www.kaggle.com/datasets/lislejoem/us-minimum-wage-by-state-from-1968-to-2017.
- [2] Wrangler, The Data. “Unemployment in America, per US State.” Kaggle, 2 Mar. 2023, www.kaggle.com/datasets/justin2028/unemployment-in-america-per-us-state.